CHAPTER 6

Bone Tissue and the Skeletal System

Bone structure, formation, fractures, and calcium balance.

Anatomy & Physiology · Broward-Miami Health Institute

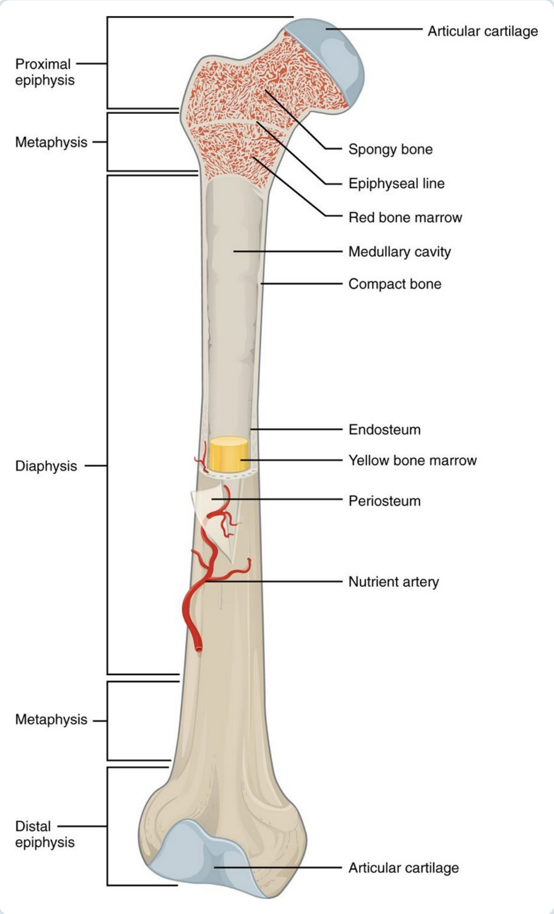
FACULTY EDITION

CHAPTER 6

Learning Objectives

After studying this chapter, students will be able to:

1. List and describe the functions of bones
2. Describe the classes of bones
3. Discuss the process of bone formation and development
4. Explain how bone repairs itself after a fracture
5. Discuss the effect of exercise, nutrition, and hormones on bone tissue
6. Describe how an imbalance of calcium can affect bone tissue Bones make good fossils



The anatomy of a long bone.

6.2

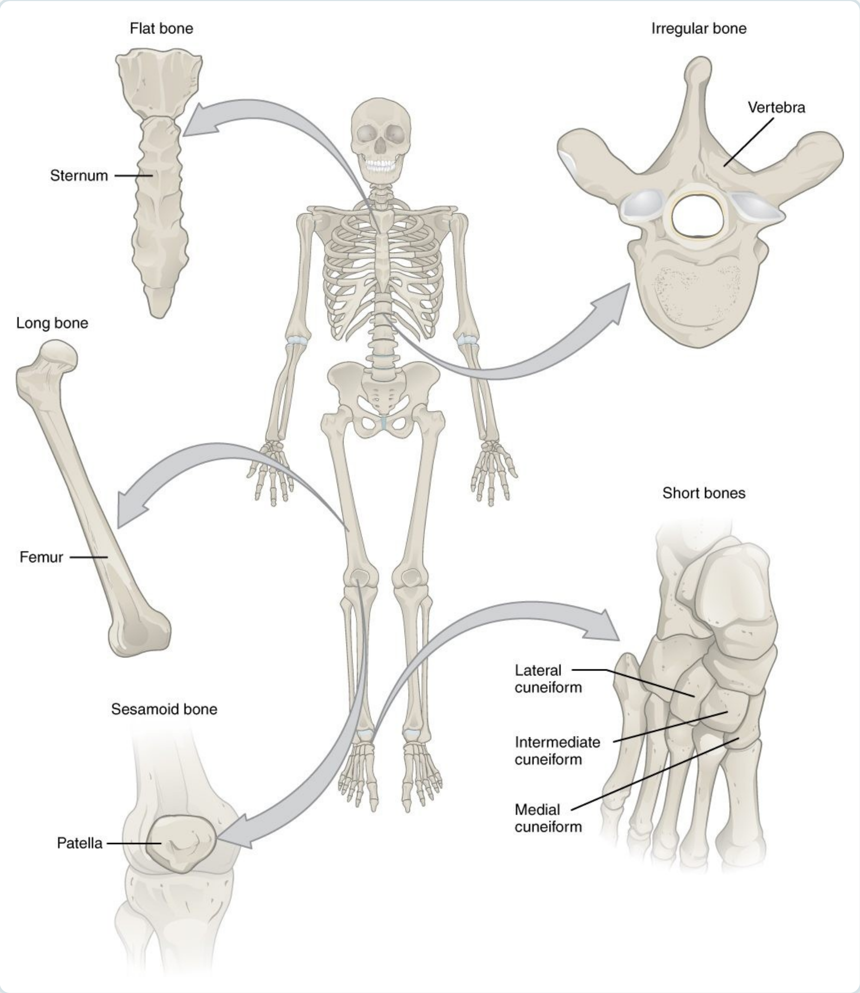
SECTION

Bone Classification

The five shapes of bones and how shape relates to function.

SECTION 6.2

Classification of Bones



The classification of bones by shape

- Long bones (limbs) act as levers
- Short bones (wrist, ankle) add stability
- Flat bones (skull, ribs) protect
- Irregular and sesamoid bones

6.3

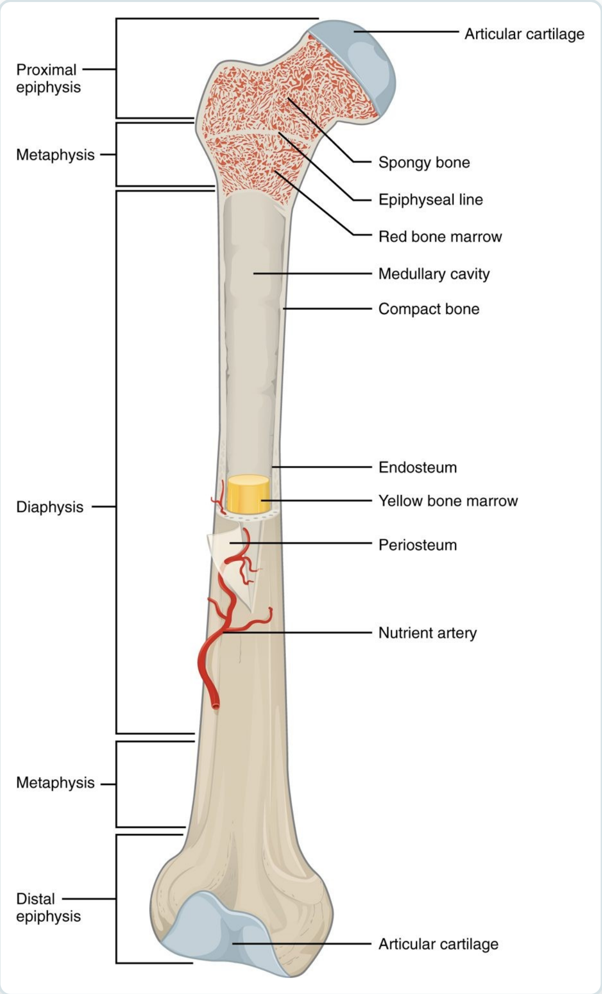
SECTION

Bone Structure

The gross and microscopic structure of bone tissue.

SECTION 6.3

Anatomy of a Long Bone

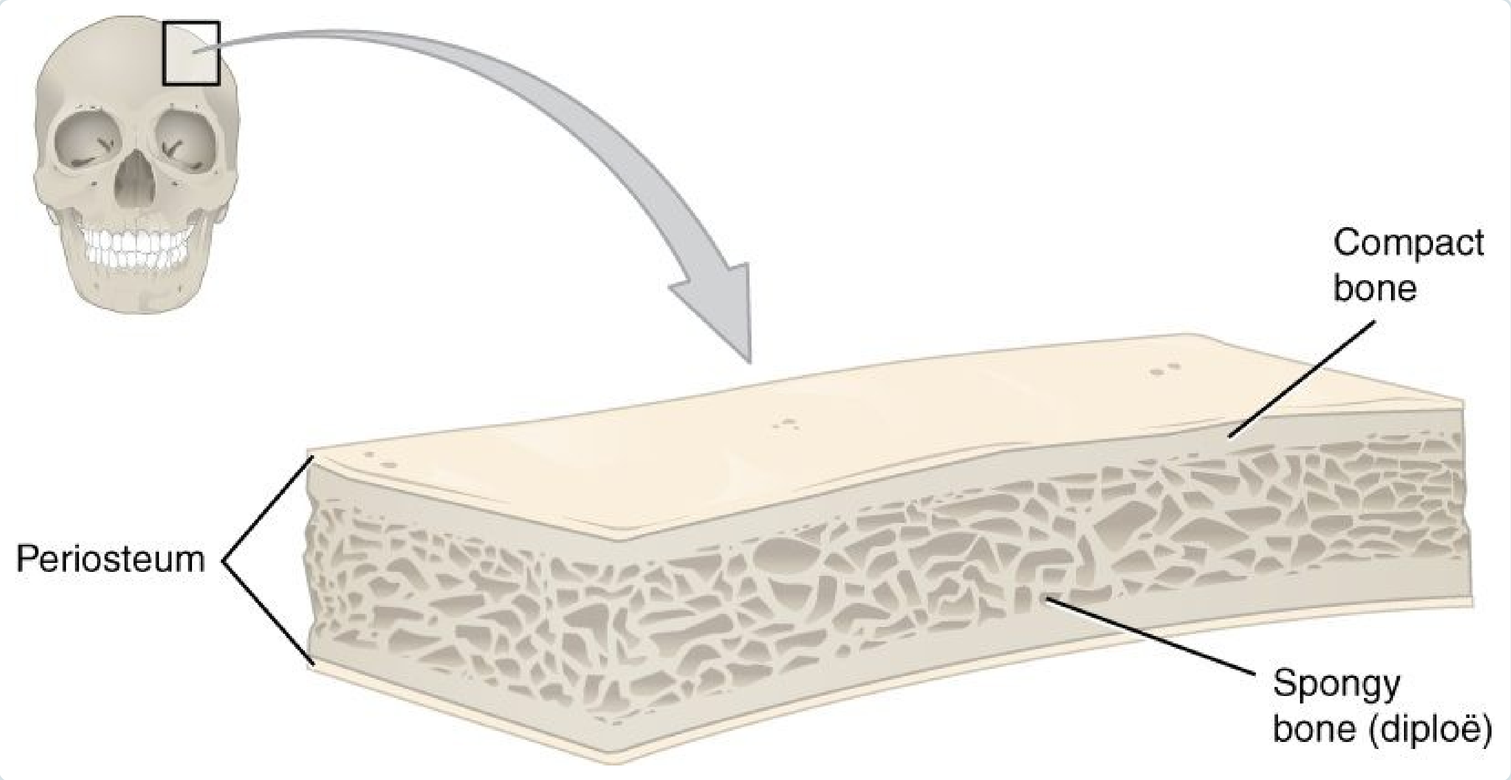


The structure of a long bone

- Diaphysis (shaft) and epiphyses (ends)
- Medullary cavity holds marrow
- Periosteum covers the outer surface
- Articular cartilage cushions the joints

SECTION 6.3

Compact and Spongy Bone



Compact and spongy bone

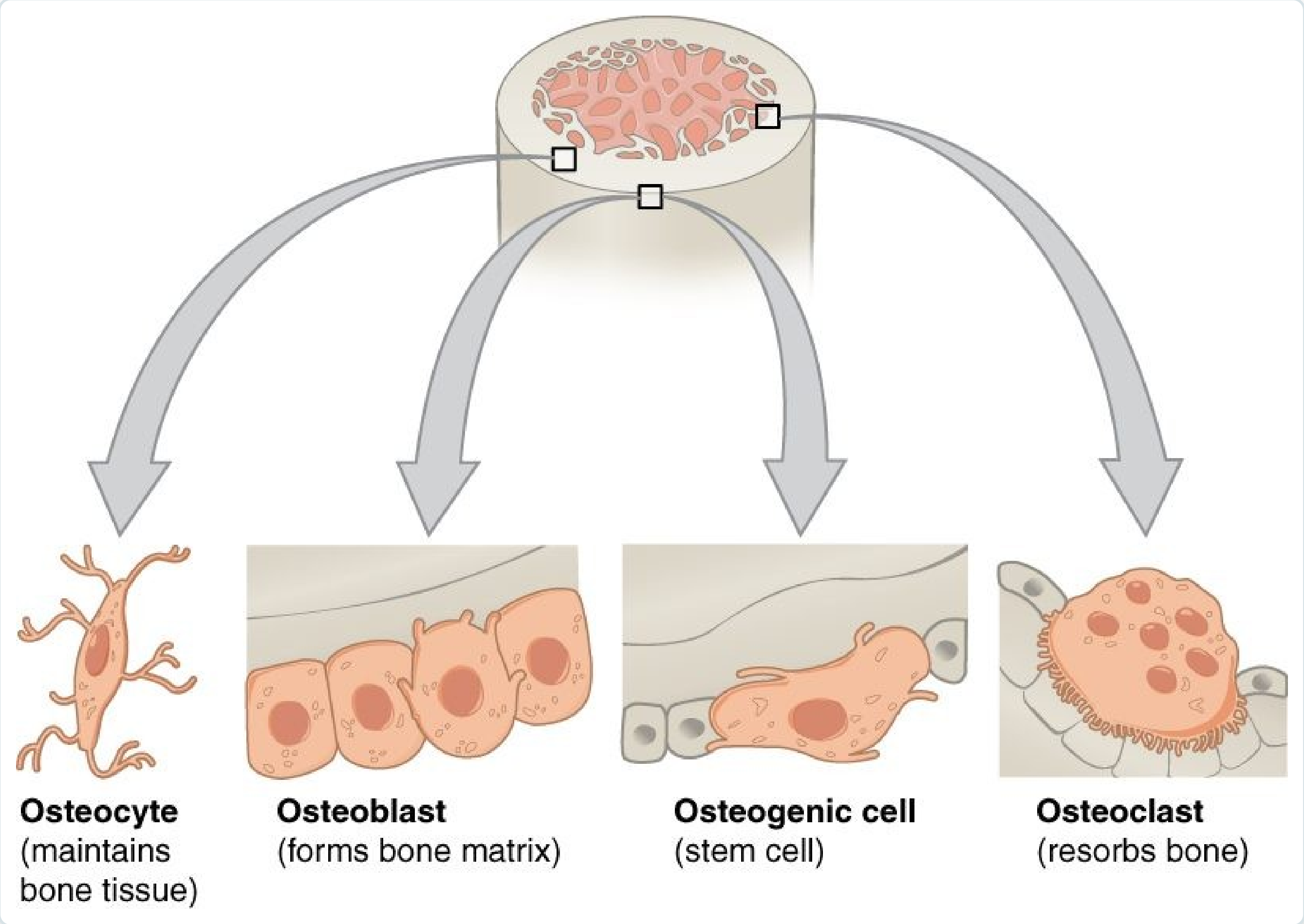
- Compact bone is dense and solid
- Forms the strong outer layer

- Spongy bone is porous (trabeculae)
- Trabeculae resist stress, hold marrow

SECTION 6.3

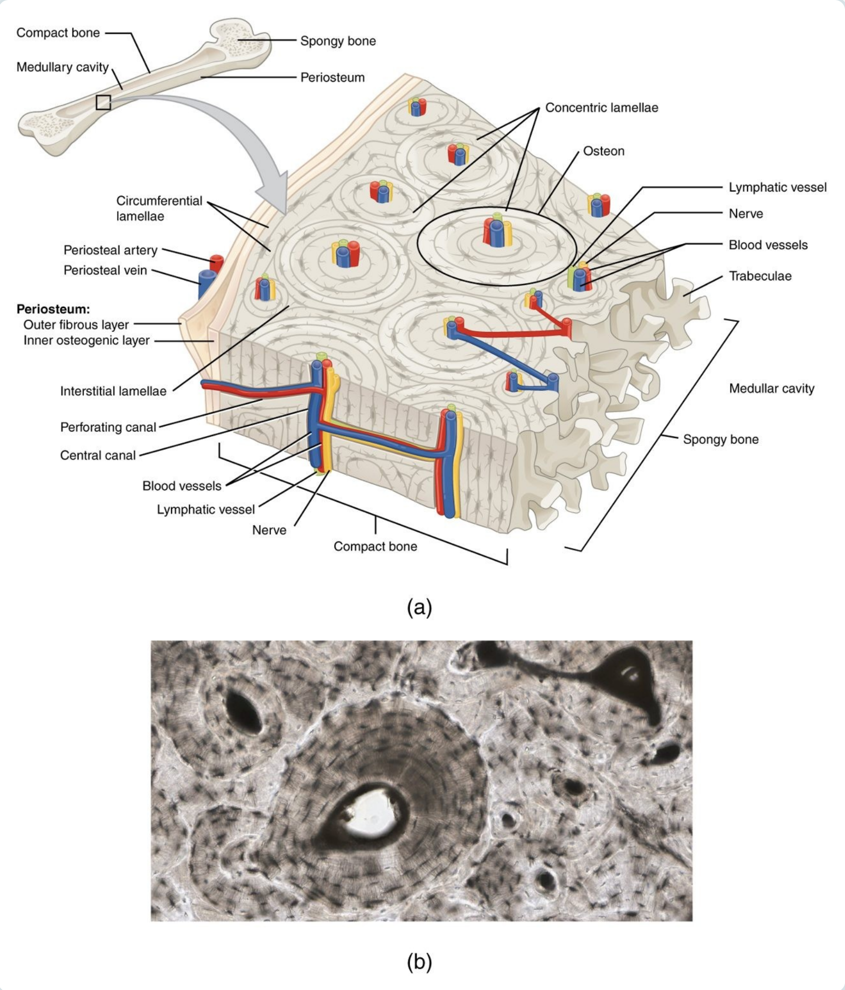
The Four Bone Cells

- Osteogenic cells are bone stem cells
- Osteoblasts build new bone
- Osteocytes maintain bone tissue
- Osteoclasts resorb (break down) bone



The cells of bone tissue

Microscopic Bone — The Osteon



The osteon (Haversian system)

- Compact bone is built of osteons
- Central canal carries vessels and nerves
- Concentric lamellae surround the canal
- Osteocytes sit in lacunae

6.4

SECTION

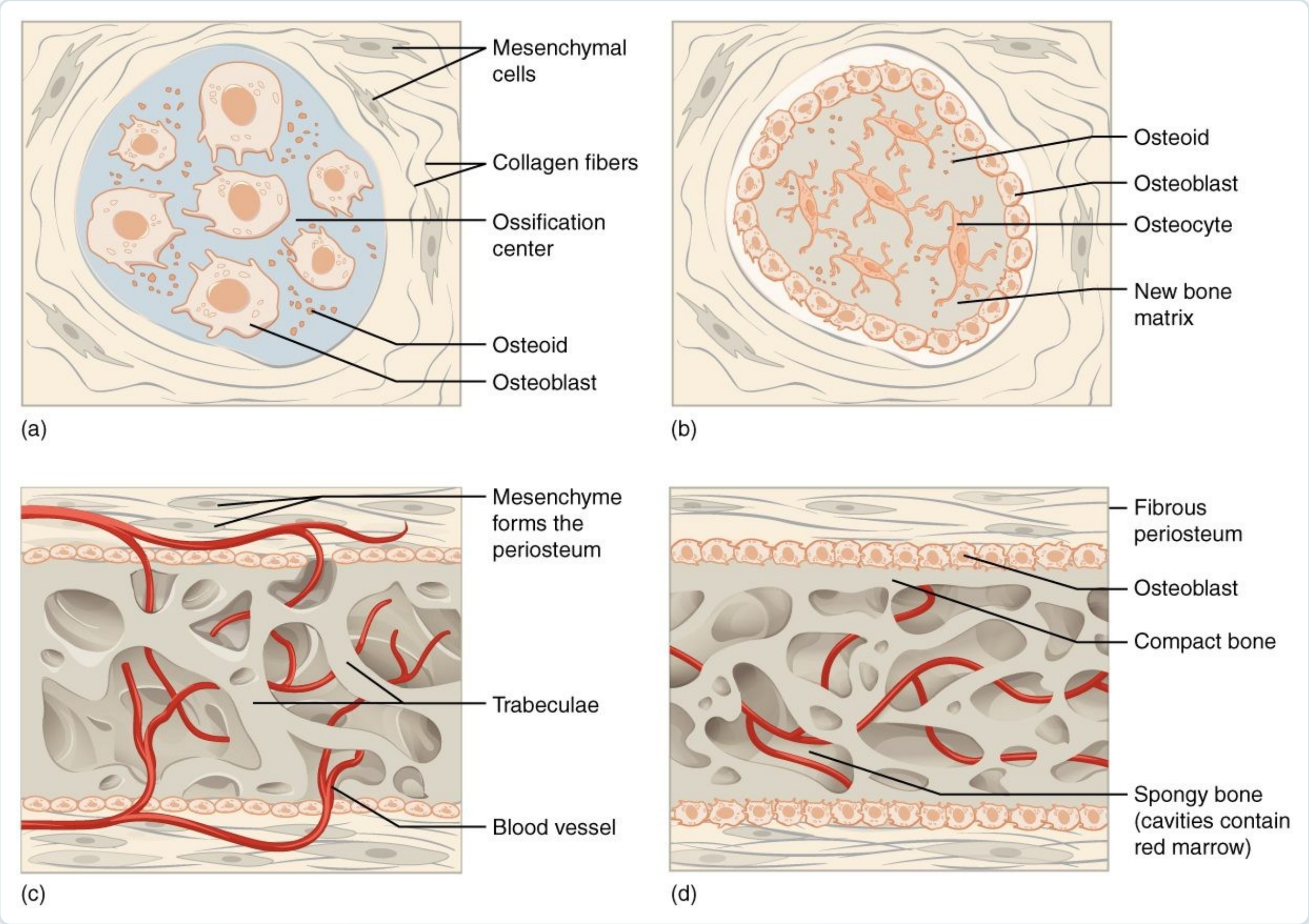
Bone Formation and Development

How bones form before birth and grow throughout childhood.

SECTION 6.4

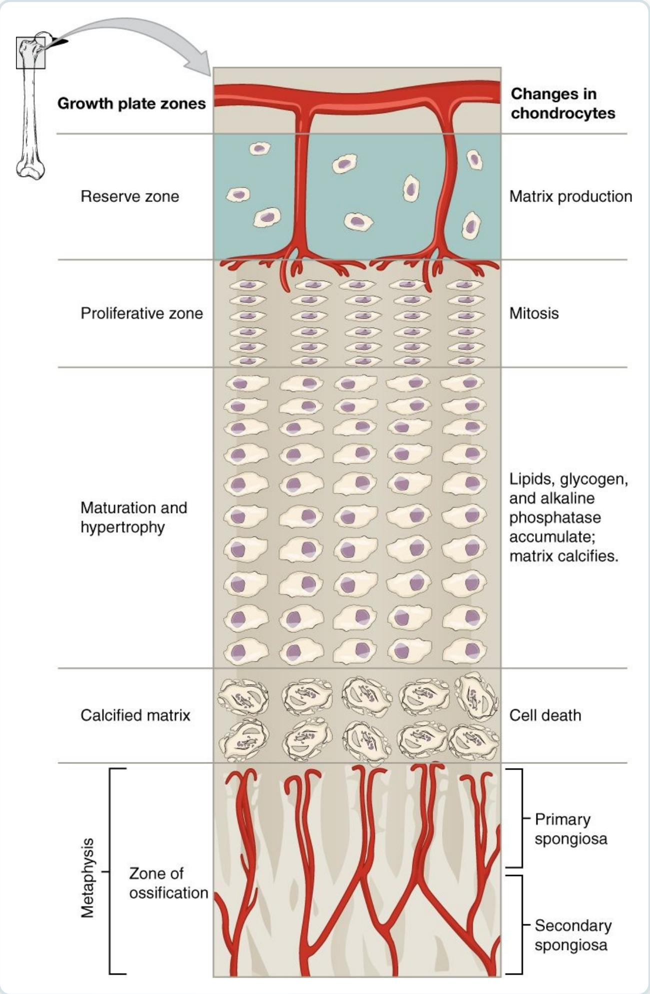
Bone Formation (Ossification)

- Intramembranous forms flat bones
- Endochondral replaces cartilage
- Most bones form from a cartilage model
- Begins before birth



Intramembranous and endochondral ossification

The Epiphyseal (Growth) Plate



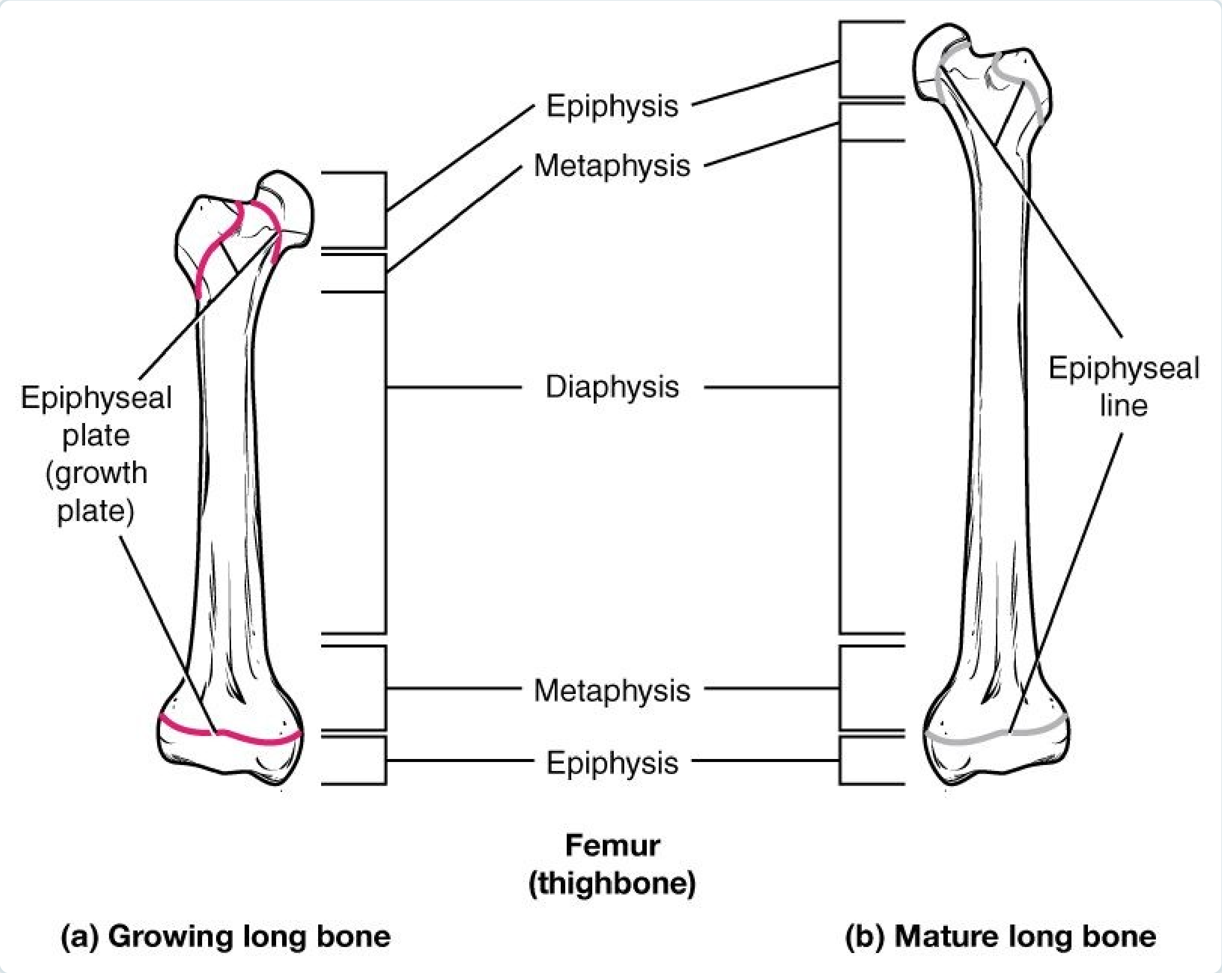
Zones of the epiphyseal plate

- A plate of hyaline cartilage
- Lengthens long bones in childhood
- Cartilage grows, then turns to bone
- Closes after puberty

SECTION 6.4

Growing vs. Mature Bone

- Active growth plate in children
- Epiphyseal line in adults
- Marks where growth occurred
- Plate injuries can affect growth



Growing and mature long bone

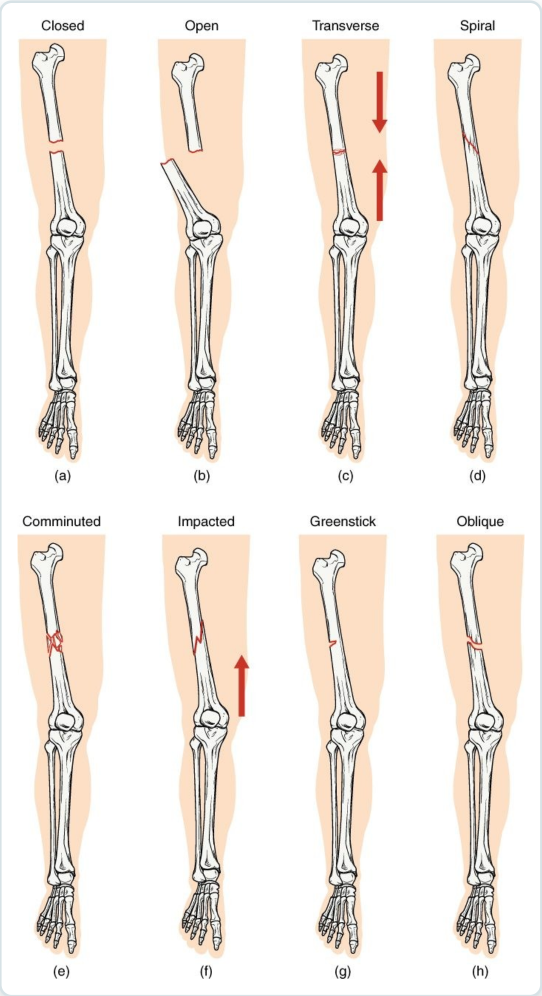
6.5

SECTION

Fractures: Bone Repair

Types of fractures and how bone repairs itself.

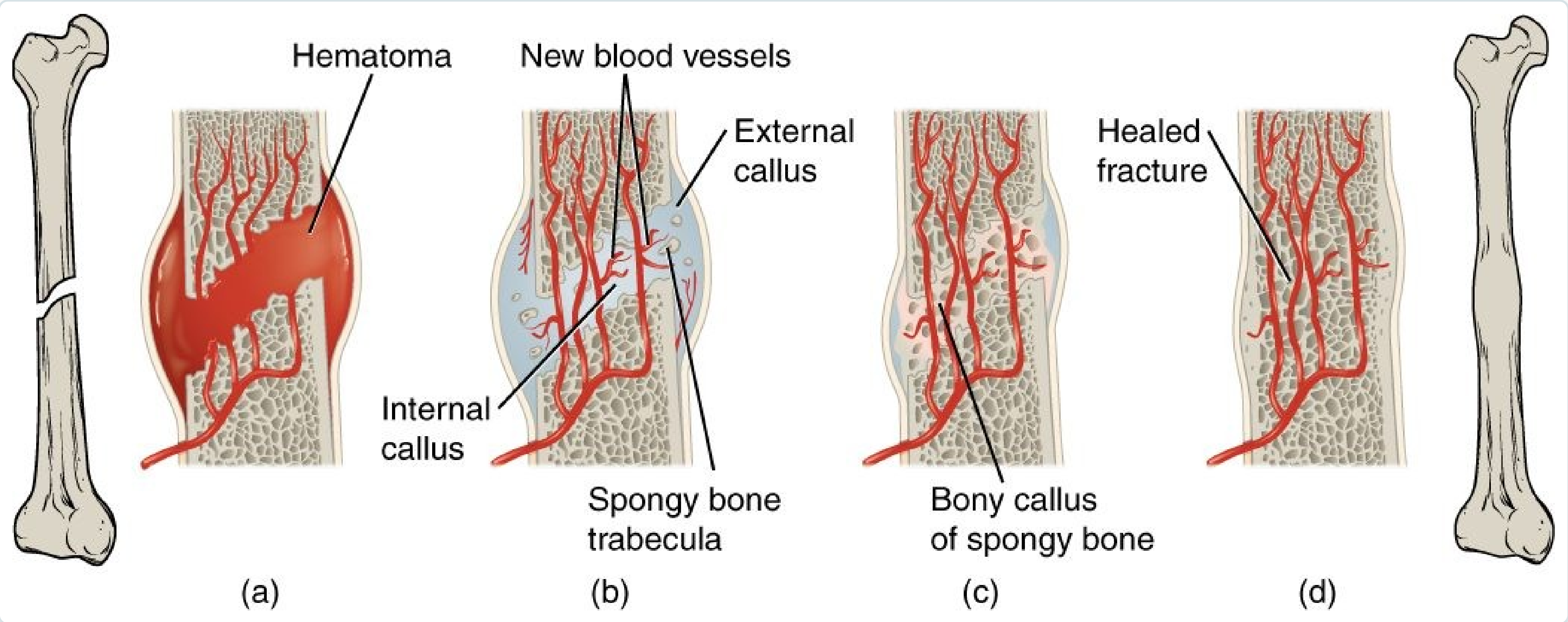
Types of Fractures



Common types of bone fractures

- Closed (simple) vs. open (compound)
- Transverse, oblique, spiral, comminuted
- Greenstick (incomplete) in children
- Impacted ends driven together

Fracture Repair



The stages of fracture repair

- A hematoma forms at the break
- A soft (fibrocartilage) callus bridges it
- A bony callus replaces it
- Remodeling restores the bone

SECTION

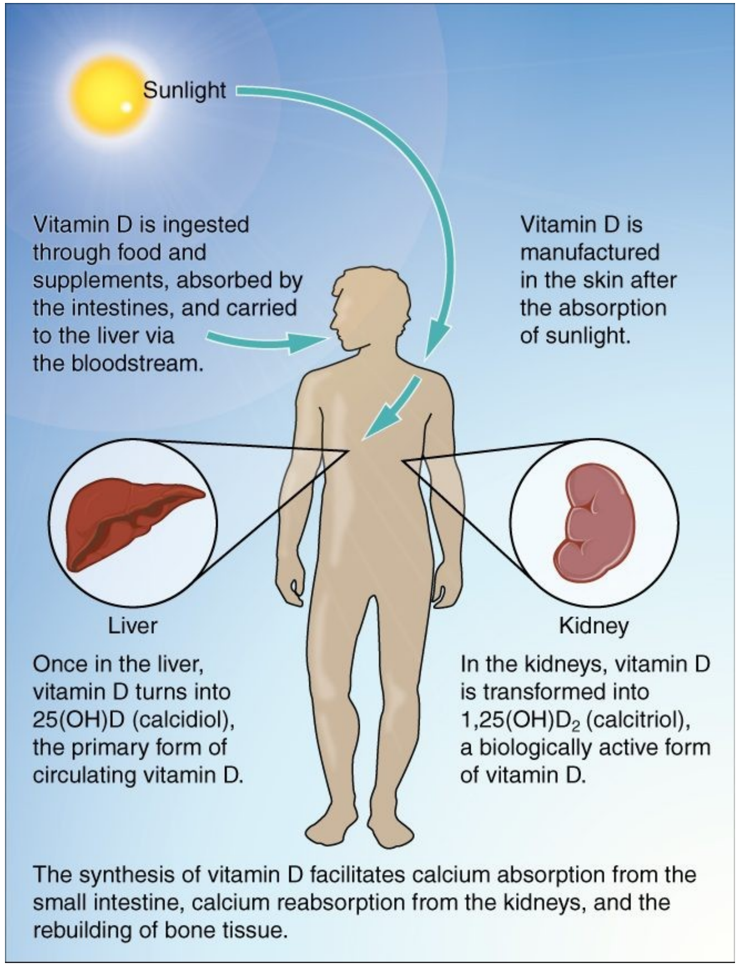
6.6

Exercise, Nutrition, Hormones, and Bone Tissue

How exercise, nutrition, and hormones keep bone strong.

SECTION 6.6

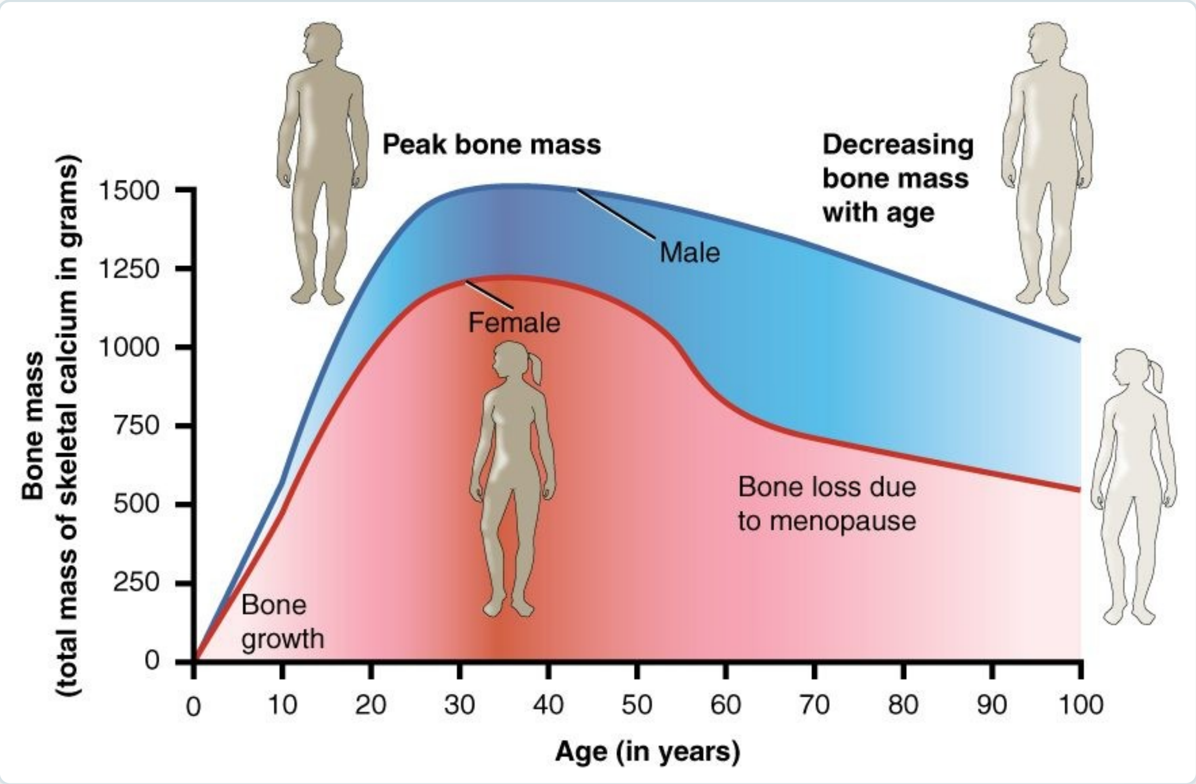
Vitamin D & Calcium Absorption



Vitamin D synthesis and calcium absorption

- Sunlight starts vitamin D synthesis
- Liver and kidneys activate it
- Vitamin D drives calcium absorption
- Essential for strong bones

Bone Mass Over the Lifespan



Bone mass across the lifespan

- Bone mass peaks in young adulthood
- Declines with age, faster after menopause
- Low bone mass leads to osteoporosis
- Exercise and calcium slow the loss

SECTION

6.7

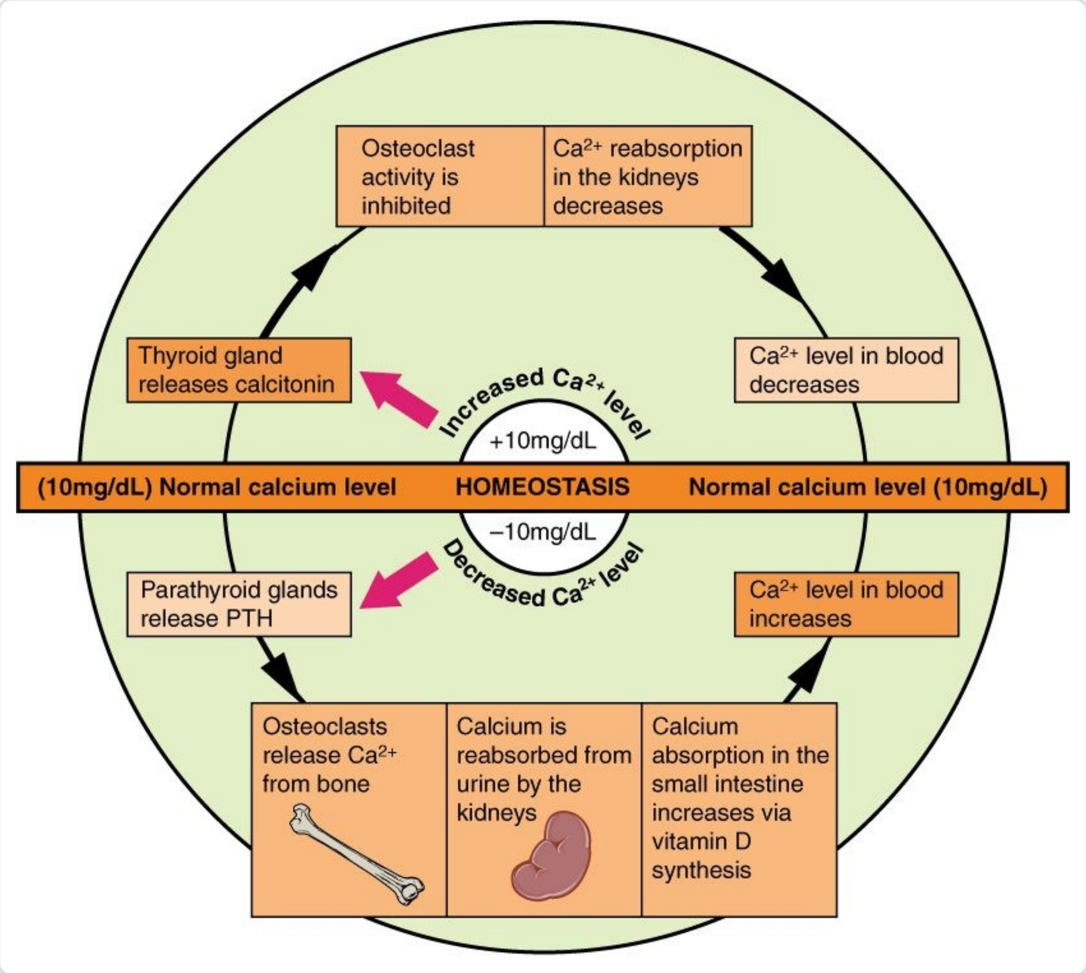
Calcium Homeostasis: Interactions of the Skeletal System and Other Organ Systems

How the skeleton helps keep blood calcium in a safe range.

SECTION 6.7

Calcium Homeostasis

- Blood calcium is tightly controlled
- PTH raises calcium (bone, kidney, gut)
- Calcitonin lowers calcium
- Bone is the body's calcium reservoir



Hormonal control of blood calcium

Key Takeaways

- The skeleton supports, protects, moves the body, stores minerals, and makes blood cells.
- Bones are classified by shape, with structure suited to function.
- Bone is living tissue, built of osteons and continuously remodeled by bone cells.
- Bones form by ossification and lengthen at the epiphyseal plate until adulthood.
- Bone health depends on exercise, nutrition, hormones, and calcium balance.

CLINICAL CONNECTIONS

Why Bone Matters in Practical Nursing

Fracture Care

Knowing fracture types and healing stages guides immobilization and recovery.

Osteoporosis

Bone loss with age raises fracture risk — central to fall prevention.

Calcium & Vitamin D

Essential nutrients for bone and for nerve and muscle function.

Blood Cell Formation

Red marrow makes the blood cells you monitor in lab work.

Growth Plates

Pediatric growth-plate injuries require special attention.

Mobility

Bone strength underpins safe movement and weight-bearing.